$$[Q, K, V] = ZU_{qkv}, \quad Q, K, V \in \mathbb{R}^{S \times d_h}$$

$$A = \mathrm{softmax}_{\mathrm{row}}(QK^{\top}/\sqrt{d_h})$$

$$\mathrm{SA}(Z) = AV, \quad A \in \mathbb{R}^{S \times S}$$